x_1	x_2	x_3	x_4	x_5	x_6
(1)	(2)	(1)	(5)	(6)	(6)
(1)	(1)	(2)	(4)	(5)	(6)
C1	C1	C1	C2	C2	C2

$\begin{vmatrix} 4 \\ -2 \\ -2 \end{vmatrix}$ *Initial* W = **REPEAT using Step η = 1**

C_1 positive C_2 negative **Zero** is unaccepted

Check classes of $Z_1 \begin{vmatrix} 0 \\ 0 \end{vmatrix}$ $Z_1 \begin{vmatrix} 7 \\ 6 \end{vmatrix}$ $Z_1 \begin{vmatrix} 3 \\ 2 \end{vmatrix}$

Ugmented Vectors (from Top)

x_1	x_2	x_3	x_4	x_5	x_6
(1)	(1)	(1)	(1)	(1)	(1)
(1)	(2)	(1)	(5)	(6)	(6)
(1)	(1)	(2)	(4)	(5)	(6)
C_1	C_1	C_1	C_2	C_2	C_2

Iteration 1

$d(x_1) = \begin{vmatrix} 1 & 1 & 1 \end{vmatrix} \cdot \begin{vmatrix} 4 \\ -2 \\ -2 \end{vmatrix} = 0$ ⇒ Update Weight $\underline{W} = \begin{vmatrix} 4 \\ -2 \\ -2 \end{vmatrix} + 1 * \begin{vmatrix} 1 \\ 1 \\ 1 \end{vmatrix} = \begin{vmatrix} 5 \\ -1 \\ -1 \end{vmatrix}$ ⇐

$d(x_2) = \begin{vmatrix} 1 & 2 & 1 \end{vmatrix} \cdot \begin{vmatrix} 5 \\ -1 \\ -1 \end{vmatrix} = 2$ ⇒ Keep same Weight $\underline{W} = \begin{vmatrix} 5 \\ -1 \\ -1 \end{vmatrix}$

$d(x_3) = \begin{vmatrix} 1 & 1 & 2 \end{vmatrix} \cdot \begin{vmatrix} 5 \\ -1 \\ -1 \end{vmatrix} = 2$ ⇒ Keep same Weight $\underline{W} = \begin{vmatrix} 5 \\ -1 \\ -1 \end{vmatrix}$

$d(x_4) = \begin{vmatrix} 1 & 5 & 4 \end{vmatrix} \cdot \begin{vmatrix} 5 \\ -1 \\ -1 \end{vmatrix} = -4$ ⇒ Keep same Weight $\underline{W} = \begin{vmatrix} 5 \\ -1 \\ -1 \end{vmatrix}$

$d(x_5) = \begin{vmatrix} 1 & 6 & 5 \end{vmatrix} \cdot \begin{vmatrix} 5 \\ -1 \\ -1 \end{vmatrix} = -6$ ⇒ Keep same Weight $\underline{W} = \begin{vmatrix} 5 \\ -1 \\ -1 \end{vmatrix}$

$d(x_6) = \begin{vmatrix} 1 & 6 & 6 \end{vmatrix} \cdot \begin{vmatrix} 5 \\ -1 \\ -1 \end{vmatrix} = -7$ ⇒ Keep same Weight $\underline{W} = \begin{vmatrix} 5 \\ -1 \\ -1 \end{vmatrix}$